ARITRA HALDER

Contact Information

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Research Interests

Bayesian Modeling, Spatial and Temporal Statistics, Statistical Computing.

Employment

Drexel University

Assistant Professor of Biostatistics. *2022–*

Adjunct, Department of Biomedical Sciences, Elkins Park, PA. *2024–*

Research Collaborator, Children's Hospital of Phila. w. Paula Magee. *2025–*

University of Virginia Research Assistant Professor *2020–2022*

Education

University of Connecticut, Storrs, Connecticut USA *July 2020*

Ph.D., M.S., Statistics

Chennai Mathematical Institute, Siruseri, Chennai India

M.Sc., Applied Mathematics *May 2016*

Presidency College, Calcutta University, Kolkata, West Bengal India

B.Sc., Honors in Statistics *June 2014*

Professional Experience

The Travelers Companies, Hartford, Connecticut, USA. 2016 – 2020

Honors and Awards

University of Connecticut

Pre-Doctoral Dissertation Fellowship:
Awarded as a part of the training grant sponsored by the Travelers Companies *2016-20*

Financial Award: *Spring 2017*
To attend SAMSI Workshop on the Interface of Statistics and Optimization.

Doctoral Dissertation Fellowship: Awarded by the Graduate School. *Spring 2020*

Doctoral Student Travel Fellowship: Awarded by the Graduate School. *Spring 2020*

IMS Hannan Graduate Student Travel Award ($500): Awarded by the Institute of Mathematical Statistics. *May 2020*

ISBA Travel Award ($300): Awarded by the International Society for Bayesian Analysis. *February 2022*

Federal Grants Received

Current

1. NSF DMS 2515897/2515899/2515898 (Role: Lead PI (33%), Collaborative Proposal with Didong Li, Co-PI, UNC and Sudipto Banerjee, Co-PI, UCLA) Rates of Change and Boundary Assessment in Spatiotemporal Processes. Project Period: 2026-2029. Amount: $320,000. PI Portion: $100,000. Start Date: June, 1, 2026
2. R21 CA305033 National Institutes of Health, National Institute of Environmental Health Sciences (Role: co-I (8%), PI: Anneclaire DeRoos) *Cancer Risks from Unregulated Contaminants of Public Water Systems.* Project Period: 2026-2027. Amount: $402,425

Past

1. R56 ES036250, National Institutes of Health, National Institute of Environmental Health Sciences (Role: co-I (17%), PI: Anneclaire DeRoos) *Extreme Weather Impacts on Children's Acute and Chronic Health Outcomes: a Multi-Site Study with Evaluation of Vulnerabilities.* Project Period: 2024-2025. Amount: $687,343
2. United States Department of Agriculture (USDA) Strategic Priority Grant Program (SPGP): *Impacts of Rural Broadband and Broadband Programs Across Racial and Ethnic Groups.* (25%) Role: (co-PI w/ Sallie Keller, University of Virginia). Amount: $600,000
3. Army Research Institute (ARI) (Grant Contract ID: W911NF-20-2-0027): *Modeling of Veteran career pathways within the Army.* (25%) Amount: $2,124,158 (co-PI w/ Joshua R. Goldstein & Sallie Keller, University of Virginia)

Courses Taught

Drexel University, Philadelphia, Pennsylvania — Fall 2022 – Present

- BST Elective: Special Topics in Statistics — Spring Qtr. 2023
- BST Elective: Spatial and Spatiotemporal Modeling with Applications — Winter Qtr. 2024
- BST: Advanced Statistical Computing — Spring Qtr. 2024–
- BST: Generalized Linear Models — Winter Qtr. 2025–
- BST: Statistical Inference I — Winter Qtr. 2026
- Urban Health Collaborative, Summer Workshop — 2024–
  Spatial and Spatiotemporal Modeling with Applications in R
- Honorary Course Offering (12 weeks) — 2024–
  Review of Statistical Inference I & II–Ph.D. Qualifying Examination Prep.
- Independent Study Victoria Ryan — Fall 2024
  Respondent Driven Sampling to Determine Network Size

University of Connecticut, Storrs, Connecticut USA

1. *Instructor* — Jan., 2020 – May, 2020
   - STAT 3025-Q Statistical Methods — Spring '20
2. *Teaching Assistant* — Aug., 2016 – Jul., 2020

   STAT 1000-Q Introduction to Statistics, STAT 1100-Q Elementary Concepts of Statistics. STAT 3494-W Seminar Writing Course in Statistics.
3. *Teaching Assistant* — Winter Intersession, 2017
   - STAT 1100-Q Introduction to Statistics.

Doctoral Students

- Daphne Liu (Expected 2029) co-advised w. Loni P. Tabb
- Emama Amin (Expected 2029) co-advised w. Loni P. Tabb
- Sabrina Afroz (Expected 2029) co-advised w. Loni P. Tabb

Masters Students

- Karlin Moore, MS, Biostatistics (2023) Thesis Title: *Spatial and Temporal Trends in Air Pollution at the State and Sub-city Level in Pennsylvania and South Philadelphia, PA (Winner of Best Thesis Award, DSPH)*
- Andrew Reid (2026)

Software Packages

1. GitHub (Non Peer Reviewed): sptwombling; spWombling; sptwdglm; manifoldGPgrad
2. CRAN (Peer Reviewed): nimblewomble (2025) Bayesian Wombling using 'nimble' [DOI]

Publications

In preparation

18. Halder, A., Li, D. and Banerjee, S., Smoothness in Scalable Gaussian Processes
17. Li, D., **Halder, A.** and Banerjee, S., On statistical inference for rates of change in spatial processes over Riemannian manifolds.
16. Chen, J., **Halder, A.**, Banerjee, S., Li, Y. and Li, D., The Nearest-Neighbor Derivative Process: Modeling Spatial Rates of Change in Massive Datasets.

Submitted

15. Palmer-McCabe, A., Jahn, J., **Halder, A.**, Pearsall, H., Kondo C., M, De Roos J., A. (2026+) Association of Racial Segregation and Violations of the Safe Drinking Water Act in the United States 2010 – 2020, *Environmental Research: Health*
14. **Halder, A.**, Stinziano, J. and Persi, J. (2025+) Bayesian Wombling for Tracking Invasive Species, *Minor revision in Environmental and Ecological Statistics*

Peer-reviewed

13. **Halder, A.** and Banerjee, S. (in press, 2025+) `nimblewomble:` An R package for Bayesian Wombling using `nimble`, *R-Journal* DOI
12. Chen, J., Xiong, C., Sun, Q., Wang, G. W., Gupta, G. P., **Halder, A.**, Li, Y. and Li, D. (in press, 2024+) Investigating spatial dynamics in spatial omics data with StarTrail, *Journal of the American Statistical Association, Applications & Case Studies.* DOI PMC1346541
11. **Halder, A.**, Li, D. and Banerjee, S. (in press, 2024+) Bayesian Spatiotemporal Wombling, *Journal of the American Statistical Association, Theory & Methods.* DOI PMC13664449
10. Charankevich, H., Goldstein, J., **Halder, A.**, Pender, J. (2023). Measuring Efficacy of the Rural Broadband Initiatives: Evidence from the Housing Market. In Proceedings of Eighth International Congress on Information and Communication Technology. ICICT 2023. *Lecture Notes in Networks and Systems, vol 693. Springer, Singapore.* DOI
9. **Halder, A.**, Mohammed, S. and Dey, D. K. Bayesian Variable Selection in Double Generalized Linear Tweedie Spatial Process Models. (2023) (in-press) *New England Journal of Statistics in Data Science: Special Issue on Modern Bayesian Methods with Applications in Data Science.* 1(2) 187–199 DOI
8. **Halder, A.**, Banerjee, S. and Dey, D. K. Bayesian Modeling with Spatial Curvature Processes. (2023) 119(546), pp. 1155–1167, *Journal of the American Statistical Association, Theory & Methods.* DOI PMC11238907

7. **Halder, A.**, Mohammed, S., Chen, K. and Dey, D. K. Spatial risk estimation in Tweedie double generalized linear models. (2022) *Proceedings of International E-Conference on Mathematical and Statistical Sciences: A Selçuk Meeting.*
6. Babu, G. R., Ray, D., Bhaduri, R., **Halder, A.** and Kundu, R. , Menon, G. I. and Mukherjee, B. (2021) COVID-19 Pandemic in India: Through the Lens of Modeling. *Global Health: Science and Practice.* DOI
5. **Halder, A.**, Mohammed, S., Chen, K. and Dey, D. K. (2021) Spatial Tweedie exponential dispersion models: an application to insurance rate-making. *Scandinavian Actuarial Journal* 10(2021) pp. 1017–1036. DOI
4. Sen, K., Mukherjee, R., Sansare, S., **Halder, A.**, Kashi, H., Ma, A. W. K., Chaudhuri, B. (2021) Impact of powder-binder interactions on 3D printability of pharmaceutical tablets using drop test methodology *European Journal of Pharmaceutical Sciences.* 160 105775. DOI
3. Ray, D., Salvatore, M., Bhattacharyya, R., Wang, L., Mohammed, S., Purkayastha, S., **Halder, A.**, Rix, A., Barker, A., Kleinsasser, M., Zhou, Y., Song, P., Bose, D., Banerjee, M., Baladandayuthapani, V., Ghosh, P., Mukherjee, B. (2020) *Predictions, role of interventions and effects of a historic national lockdown in India's response to the COVID-19 pandemic: data science call to arms. Harvard Data Science Review, Special Issue-1.* DOI
2. Mukherjee, R., **Halder, A.**, Naik, S., Sansare, S. and Chaudhuri, B. (2020). Statistical quantification of parameters altering tribo-charging in pharmaceutical mixtures. *Featured Article, Journal of Pharmaceutical Sciences.* DOI
1. Das, S., **Halder, A.**, and Dey, D. K. (2017). Regularizing Portfolio Risk: A Bayesian Approach. *Methodology and Computing in Applied Probability.* 19(3) 865–889. DOI

Non-Peer Reviewed

1. Das, S., **Halder, A.**, Lahiri, A. and Dey, D. K. (2018) Modeling portfolio risk and return using Dirichlet process prior. arXiv

Conference Abstracts

1. Sharma, R., Frueh, L., Joyce, E. K., Just, A. C., **Halder, A.**, Sheffield, P. E., and Clougherty, J. (2024) Spatial heterogeneity in the acute effects of ambient air pollution and temperature on pediatric seizures and epilepsy across New York, *ISEE Conference Abstracts.* DOI

Departmental Service

Committees

- Co-chair EPI/BIO Seminar Committee *2022–2025*
- Biostatistics Program Committee *2023–Present*
- Open-Rank Biostatistics Faculty Search Committee *2024–2025*

Doctoral Dissertation Committees

- Jihyeon Kwon, PhD. Candidate, Biostatistics. Dissertation Advisor: Harrison Quick.
- Victoria Ryan, PhD. Candidate, Epidemiology. Dissertation Advisor: Ayden Scheim.
- Amber Palmer, PhD. Candidate, Epidemiology. Dissertation Advisor: Anneclaire DeRoos.
- Apollo Kivumbi, PhD. Candidate, Epidemiology. Dissertation Advisor: Kristen Lyall.
- Jenna Carcione, PhD. Candidate, Biostatistics. Dissertation Advisor: Lucy Robinson.

- Katherine Roshak, PhD. Candidate, Epidemiology. Dissertation Advisor: Theresa Sweet.
- Deepika Chauhan, PhD. Candidate, Epidemiology. Dissertation Advisor: Brian Lee.
- Gianni Anfuso, PhD. Candidate, Biostatistics. Dissertation Advisor: Usama Bilial and Harrison Quick.
- Md. Karimuzamman, PhD. Candidate, Biostatistics. Dissertation Advisor: Brisa N. Sanchez.
- Hal Chen, PhD. Candidate, Biostatistics. Dissertation Advisor: Brisa N. Sanchez.

Academic Service

Grant Reviewer (No. of Panels: 2)

- NIH-NICD Study Section on Neuro Informatics, Computational and Data Analysis (2 panels) *2025–present*

Reviewer (Total: 30)

- R-Journal (1) *2023–present*
- Annals of Applied Statistics (5) *2023–present*
- Spatial Statistics (3) *2019– present*
- Sankhyā B (3) *2019– present*
- Journal of Spatial Science (2) *2021– present*
- Environmental and Ecological Statistics (1) *2022– present*
- Statistics in Medicine (2) *2022– present*
- Electronic Journal of Statistics (2) *2022– present*
- Journal of Royal Statistical Society: Series C (2) *2022– present*
- Journal of Urban Health (2) *2025– present*
- ASTIN Bulletin (1) *2023–Present*
- Statistics and Data Science in Imaging (1) *2026–Present*
- Scientific Reports, Nature (2) *2026–Present*
- Biostatistics (1) *2026–Present*
- Spatial and Spatiotemporal Epidemiology (2) *2026–Present*

Chairing

- IISA 2021 Conference
  - IS-58 Contemporary Bayesian Modeling with Applications
- New England Statistical Symposium, 2022
  - IS-13: Advances in Statistical Modeling of Complex Structured Brain Data
- 64-th International Statistics Institute (ISI) World Statistics Congress, Ottawa, Canada, 2023
  - Modern Advances in Statistical Analysis of Spatial and Spatiotemporal Data

Organizing

- Joint Statistical Meetings, 2022: Modern Advances in Modeling complex Spatial data
- 64-th International Statistics Institute (ISI) World Statistics Congress, Ottawa, Canada, 2023: Modern Advances in Statistical Analysis of Spatial and Spatiotemporal Data
- International Society for Bayesian Analysis World Meeting, 2024, Venice, Italy: Innovations in Spatial and Spatiotemporal Statistics and its Applications
- New England Statistical Symposium, 2024, Storrs, Connecticut: Statistical Inference, Geometry and Artificial Intelligence

Invited Department Talks and Colloquiums

- Department of Statistics, Presidency University, Kolkata, India. Invited by Prof. Biswajit Roy, Chair, Department of Statistics.
- Department of Statistics and Data Science, University of Cincinnati, Cinncinati, Ohio. Invited by Dr. Xia Wang, Associate Professor, Department of Statistics and Data Science.
- Department of Data Science, Chennai Mathematical Institute, Chennai, India. Invited by Dr. Sourish Das, Associate Professor, Department of Data Science (Apr. 22, 2022)
- Department of Biostatistics, Dornsife School of Public Health, Drexel University, Philadelphia, Pennsylvania. Invited by Dr. Leslie A. McClure, Professor and Chair (Jun. 15, 2022)
- Mathematical Biocomplexity, Department of Mathematics, University of Virginia, Charlottesville, Virginia. Invited by Dr. Christian Reidys, Professor and Director (Aug. 12, 2022)
- Department of Statistics, Iowa State University, Ames, Iowa. Invited by Dr. Sarah Nusser Emeritus Professor and Dr. Zhenguan Zhu, Professor (Sep. 12, 2022).
- Department of Statistics, George Washington University, Washington D.C. Invited by Dr. Subrata Kundu, Professor and Chair (Sep. 16, 2022).
- Department of Statistics, George Mason University, Fairfax, Virginia. Invited by Dr. Abolfazl Safikhani, Assistant Professor (Mar. 10, 2023).
- Webinar organized by ASA Philadelphia Chapter, Philadelphia, Pennsylvania. Invited by Fengqing (Zoe) Zhang, Associate Professor (Jun. 6, 2024)
- Department of Biostatistics, Pennsylvania State University, Hershey, PA. Invited by Dr. Shouhao Zhao, Professor, Department of Biostatics (Sep. 15, 2026)

Conferences & Symposiums

- 31st New England Statistics Symposium, *2017*
  Presented *Modeling Large Insurance Claims and Occurence Data.*
- Joint Statistical Meetings *2020*
  Presented (Contributed Session) *Curvature processes: Directional concavity in Gaussian random fields.*
- Conference organized by the International Indian Statistical Association (IISA) *2020*
  Presented (Contributed Session) *Spatial Tweedie exponential dispersion models: an application to insurance rate-making.*
- Joint Statistical Meetings *2021*
  Presented (Contributed Session) *Bayesian spatiotemporal modeling of Army Veteran Career Pathways*
- Federal Committee on Statistical Methodology Research and Policy Conference (Virtual) *2021*
  Creating Policy Tools for Broadband Subsidy programs: Combining Spatial Regression Discontinuity Designs and Bayesian Wombling
- EAC-ISBA *2021*
  Presented (Poster Session) *Curvature Processes: Directional Concavity in random fields*

- IISA Mini Conference (Contributed) (Virtual) *2022*
  Presented *Curvature Processes: Directional Concavity in random fields*
- Workshop in Probability and Statistical Methods, Brazil (Invited) (Virtual) *2022*
  Presented *Curvature Processes: Directional Concavity in random fields*
- 35-th New England Statistical Symposium, Storrs (Invited) (In-person) *2022*
  Presented *Bayesian Variable Selection in Double Generalized Linear Tweedie Spatial Process Models.*
- 5-th International Conference on Econometrics and Statistics (Invited) (Virtual) *2022*
  Presented *Bayesian Variable Selection in Double Generalized Linear Tweedie Spatial Process Models.*
- 24-th International Conference on Computational Statistics (Contributed) (Virtual) *2022*
  Presented *Bayesian Variable Selection in Double Generalized Linear Tweedie Spatial Process Models.*
- International E-Conference on Mathematical and Statistical Sciences: A Selçuk Meeting (Contributed) (Virtual) *2022*
  Presented *Spatial risk estimation in Tweedie double generalized linear models.*
- 10-th International Purdue Symposium on Statistics—Demystifying Data Science via Statistics: Theory and Applications, June 6–9. Session on "Emerging Challenges in Spatial Statistics". Presented *Bayesian Modeling with Spatial Curvature Processes.* *2023*
- 37-th New England Statistical Symposium, Storrs (Invited) (In-person). Session on "Advanced Spatial Learning Methods for Biomedical Applications". Presented *Bayesian Modeling with Spatial Curvature Processes.* *2024*
- Applied Statistics Symposium at Vanderbilt University, International Chinese Statistical Association (Invited) Presented *Bayesian Modeling with Spatial Curvature Processes.* *2024*
- International Chinese Statistical Association, Annual USA Chapter, UConn (Invited) Presented *Bayesian Spatiotemporal Wombling.* *2025*
- EnviBayes, Texas A&M (Invited) Presented *Bayesian Spatiotemporal Wombling.* *2025*
- National Institute of Statistical Sciences, Statistics Serving Societ (S3) Ingram Olkin Forums, *Geospatial Data for Maternal Health* *2026*

Workshops:

- SAMSI Workshop in the Interface of Statistics and Optimization (WISO), Duke University *2017*
- Broadening Participation: 2022 Mathematical and Physical Sciences (MPS) Workshop for Young Investigators, National Science Foundation (NSF) and University of Florida *2022*

Professional Society Membership

- American Statistical Association (ASA)
- Institute of Mathematical Statistics (IMS)
- International Society for Bayesian Analysis (ISBA)
- International Chinese Statistical Association (ICSA)
- International Indian Statistical Association (IISA) (Life-Member)

Academic References

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